

$$C_n = \left(\frac{C_{n,\beta}}{C_{Y,\beta}} \right)_{\text{ref}} \left(C_S \frac{u}{V} - C_D \frac{v}{V} \right) + (C_{n,\bar{p}}/C_L)_{\text{ref}} C_L \bar{p} + C_{n,\bar{r}} \bar{r} + C_{n,\delta_a} \delta_a + C_{n,\delta_r} \delta_r$$

$$T = \tau(\rho/\rho_0)^a (T_0 + T'V + T''V^2)$$

$$V = \sqrt{u^2 + v^2 + w^2}, \quad \alpha = \tan^{-1}(w/u), \quad \beta = \tan^{-1}(v/u)$$

Modify the flight simulation program written for problem 11.22 so that it uses these relations to compute the forces and moments acting on the aircraft. Use the relations developed in problem 11.23 to specify the initial condition.

- 11.25. Modify the flight simulation program written for problem 11.24 so that the lift, drag, and pitching moment coefficients are computed to incorporate the effects of stall in some manner. At angles of attack below some critical value the forces and moments should be computed from the relations given in problem 11.24. At larger angles of attack, the lift and drag computations should be changed to use the algorithm that you develop. Be sure that the relations are at least first-order continuous at all angles of attack.
- 11.26. Starting with Eqs. (11.6.3), (11.6.13), and (11.7.8), develop the relation given in Eq. (11.12.38) for the change in the Euler-Rodrigues quaternion that results from the geodesic change in bearing associated with the transformation from local flat-Earth coordinates to geographic coordinates.
- 11.27. Modify the flight simulation program that was written for problem 11.25 so that the aircraft position is described in terms of Earth-fixed latitude, Φ , longitude, Ψ , and geometric altitude, H . Use the spherical-Earth approximation specified by Eqs. (11.12.37) and (11.12.38).